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Functional Elbow Range of Motion for Contemporary Tasks

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Investigation performed at Shriners Hospitals for Children, Salt Lake City, and the Department of Orthopaedics, University of Utah, Salt Lake City, Utah

Background: Elbow range of motion for functional tasks has been previously studied. Motion arcs necessary to complete contemporary tasks such as using a keyboard or cellular telephone have not been studied and could have implications on what is considered to be a functional arc of motion for these tasks. The purpose of this study was to determine elbow range of motion, including flexion-extension, pronation-supination, and varus-valgus angulation, with use of three-dimensional optical tracking technology for several previously described positional and functional tasks along with various contemporary tasks.

Methods: Twenty-five patients performed six positional and eleven functional tasks (both historical and contemporary). Elbow flexion-extension, varus-valgus, and forearm rotation (pronation and supination) ranges of motion were measured.

Results: Positional tasks required a minimum (mean and standard deviation) of 27° ± 7° of flexion and a maximum of 149° ± 5° of flexion. Forearm rotation ranged from 20.0° ± 18° of pronation to 104° ± 10° of supination. Varus and valgus angulations ranged between 2° ± 5° of varus to 9° ± 5° of valgus. For functional tasks, the maximum flexion arc was 130° ± 7°, with a minimum value recorded as 23° ± 6° and a maximum value recorded as 142° ± 3°. All of these were for the cellular telephone task. The maximum pronation-supination arc (103° ± 34°) was found with using a fork. Maximum pronation was found with typing on a keyboard (65° ± 8°). Maximum supination was found with opening a door (77° ± 13°). Maximum varus-valgus arc of motion was 11° ± 4°. Minimum valgus (0° ± 6°) was found with cutting with a knife, while maximum valgus (13° ± 6°) was found with opening a door.

Conclusions: Functional elbow range of motion necessary for activities of daily living may be greater than previously reported. Contemporary tasks, such as using a computer mouse and keyboard, appear to require greater pronation than other tasks, and using a cellular telephone usually requires greater flexion than other tasks.

Normal motion of the elbow was first studied with use of a standard hand goniometer[1]. Simultaneous measurement of motion about the elbow and forearm with accuracy was not possible with this device despite various efforts in modifying the experimental design[2-4]. Morrey et al., in 1981, reported functional elbow range of motion with use of a triaxial electrogoniometer[5]. In their series of thirty-three subjects, they described a functional motion arc of 100°. Most activities of daily living then required from 30° to 130° of flexion and 50° of pronation to 50° of supination. Since that time, various three-dimensional technologies have evolved to measure joint motion. Electromagnetic and optical tracking systems were first used to determine various motions in the lower extremity during studies of gait patterns. More recently, these techniques have been used in the upper extremity, particularly in the measurement of elbow range of motion. Despite these advancements, replication of the original work of Morrey et al. with use of three-dimensional imaging and the reporting of varus and valgus angulations is still lacking[6-11].

The purpose of this study was to determine the functional range of motion of the elbow and forearm with use of a three-dimensional optical tracking system to examine the tasks described by Morrey et al.[5] as well as several contemporary tasks. Varus-valgus angulation arcs for these tasks were also determined.

Materials and Methods

Subjects

The study was approved by the institutional review board Human Subjects Committee at the University of Utah, Salt Lake City, Utah. A sample size of

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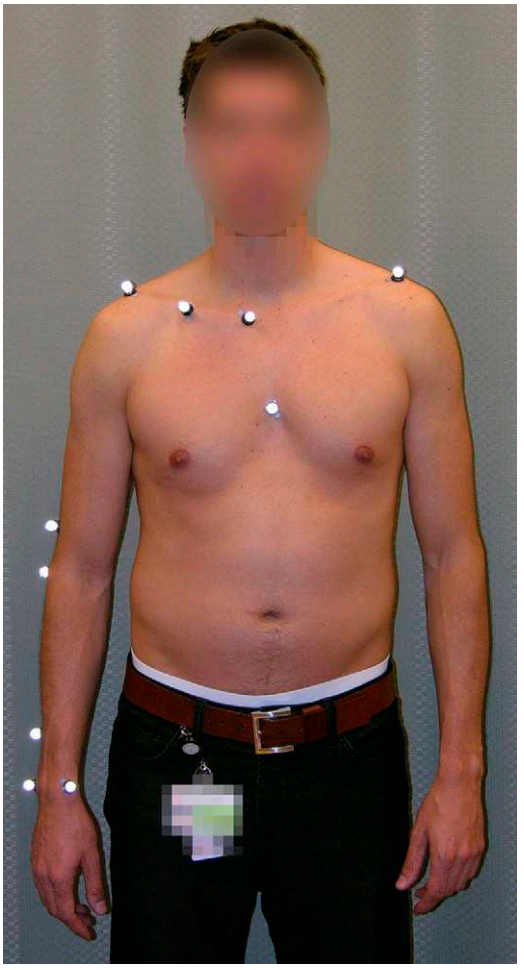


Fig. 1

Skin marker setup for each subject. Markers not shown include C7 and T10. The medial epicondyle marker was also not visualized.

convenience was used. Twenty-five healthy adult subjects (fourteen male and eleven female) participated. The average age (and standard deviation) was 34 ± 10 years. Subjects were excluded on the basis of their medical history. Radiographs were not utilized. Dominant elbows were tested for all subjects. Twenty-four right elbows and one left elbow were tested.

Instrumentation

Two generations of a ten-camera, optical, three-dimensional motion analyzer (Vicon Motion Systems, Centennial, Colorado) were used in a single laboratory, as data collection for the study spanned a system upgrade in combination with a purpose-built upper extremity model. Sampling frequency of the cameras was 60 or 100 Hz. Vicon and Nexus software (both from Vicon) were used for data assimilation, and a custom model was used for analysis. Performance and accuracy of the Vicon system has been confirmed previously[12,13]. A Cardan angle formulation with use of a flexion, abduction (valgus), and rotation (supination) sequence was used to describe elbow angles on the basis of the orientation of the forearm segment with respect to the humeral segment.

Procedures

Light-reflective spherical markers (14 mm in diameter) were attached to the skin of each subject. Markers were placed on each subject at the radial and ulnar styloids, medial and lateral humeral epicondyles, distal one-third of the radial shaft, distal one-third of the humeral shaft laterally, incisura jugularis sterni, xiphoid process, spinous process of C7, spinous process of T10, the acromioclavicular joint ipsilateral and contralateral to the elbow being measured, and the midpoint of the clavicle ipsilateral to the elbow being measured[14] (Fig. 1). A single evaluator placed all markers for all study participants. Functional motions of the shoulder and elbow were used to calculate the glenohumeral joint center and the location and orientation of the elbow flexion axis. With use of functional joint centers to establish anatomic axes of motion, error attributable to skin motion is minimized. Flexion accuracy of the model has been reported as within 2°, and repeatability of the model has been reported as within 6° for all motions[14].

Next, all twenty-five subjects were evaluated for elbow range of motion while performing six positional and eleven functional tasks. The six positional tasks included touching the vertex of the head, the occiput, chest, neck, sacrum, and finally the shoes. All subjects were instructed to start the motion from the anatomic position (defined as standing with the arms extended at each side with the palms facing anteriorly). End points of motion were recorded for flexion, varus or valgus as determined by the humerus-forearm angle in the frontal plane, and supination or pronation. Subjects were then seated in front of a desk of standard height (75 cm) (Fig. 2). They were asked to place their hands on the armrest of the chair at the beginning of each motion and then perform the eight tasks previously described by Morrey et al. as well as three contemporary tasks[5]. After the task was demonstrated by one of us (M.S.), each task was performed once by the subject. The eight previously described tasks included pouring from a pitcher into a glass, drinking from a glass, eating with a fork, cutting with a knife, reading a magazine, picking up a telephone, standing up from the desk, and opening a door[5]. The three contemporary tasks evaluated included using a standard computer mouse across a mouse pad, typing on a standard computer keyboard, and picking up from the desk and holding a cellular telephone to the ear while standing. Motion arcs for these tasks were recorded once again for flexion, varus-valgus, and supination-pronation. Motion for each task was recorded once the task was initiated to completion of the task.

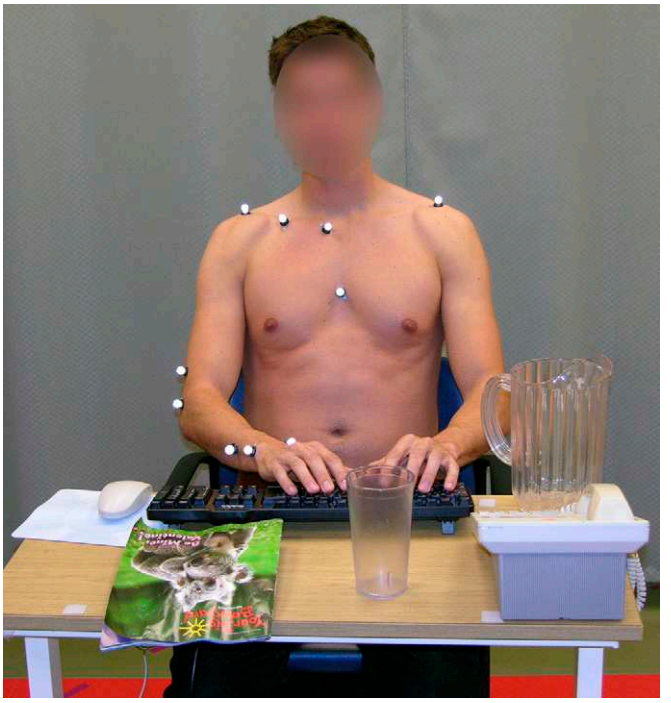


Fig. 2

Test setup for activities of daily living. Subjects are seated at a desk of standard height.

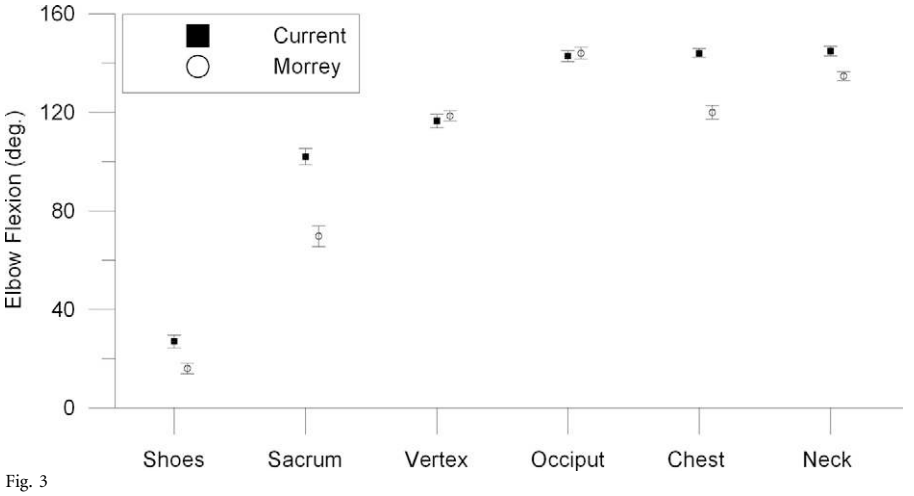


Fig. 3

Positioning test end points of motion for elbow flexion ordered from minimum to maximum. Current data are compared with those in study by Morrey et al.[5]. Error bars represent 95% confidence interval.

Data Analysis

Mean, standard deviation, and confidence interval measures were made for each positional task analyzed. Minimum, maximum, and range of motion arc measurements were made for each functional task along with confidence intervals.

Source of Funding

External funding did not play a role in the development of this study.

Results

Positional range of motion results are shown in Figures 3, 4, and 5. The minimum flexion (mean and standard deviation) required was 27° ± 7°, which was found with reaching to tie a shoe. Maximum flexion was found in activities required to reach the neck, chest, and occiput. These measured 149° ± 5°, 144° ± 5°, and 143° ± 6°, respectively. Maximum valgus positioning of the elbow was 9° ± 5°, which occurred when reaching for the vertex of the head, and maximum varus positioning was 2° ± 5°, which occurred when reaching for the sacrum. Forearm rotation ranged from 20° ± 18° of pronation in reaching for the occiput to 104° ± 10° of supination in reaching for the sacrum.

The results of functional tasks are reported as arcs of motion or as minimum and maximum values. Results are shown in Figures 6, 7, and 8. The use of a cellular telephone required the greatest flexion arc of motion (130° ± 7°). Maximum flexion recorded for the cellular telephone task was 147° ± 3°, which did not differ significantly from maximum flexion with use of a standard telephone (146° ± 3°). The largest pronation-supination arc of motion (103° ± 34°) was found with use of a fork. Maximum supination was found with opening a door and was recorded as 77° ± 13°. Pronation was greatest for the keyboard task and measured 65° ± 8°. Varus

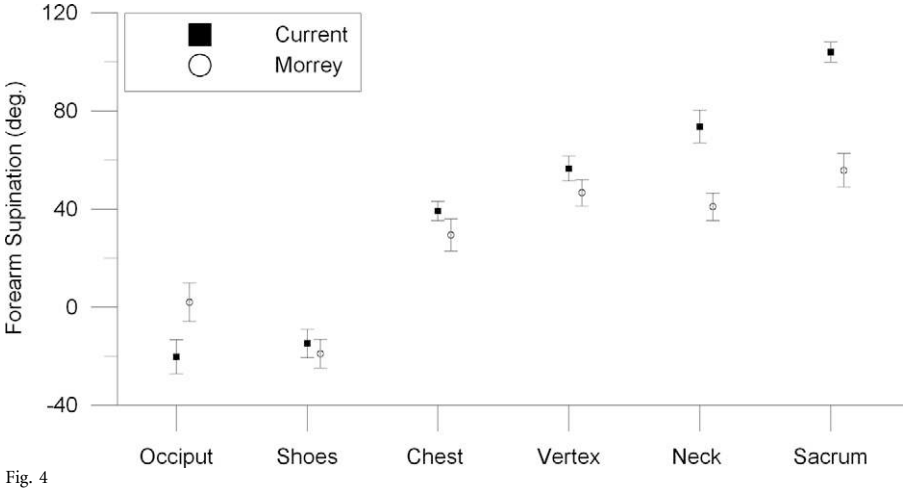


Fig. 4

Positioning test end points of motion for forearm rotation ordered from maximum pronation (negative) to maximum supination (positive). Current data are compared with those in the study by Morrey et al.[5]. Error bars represent 95% confidence interval.

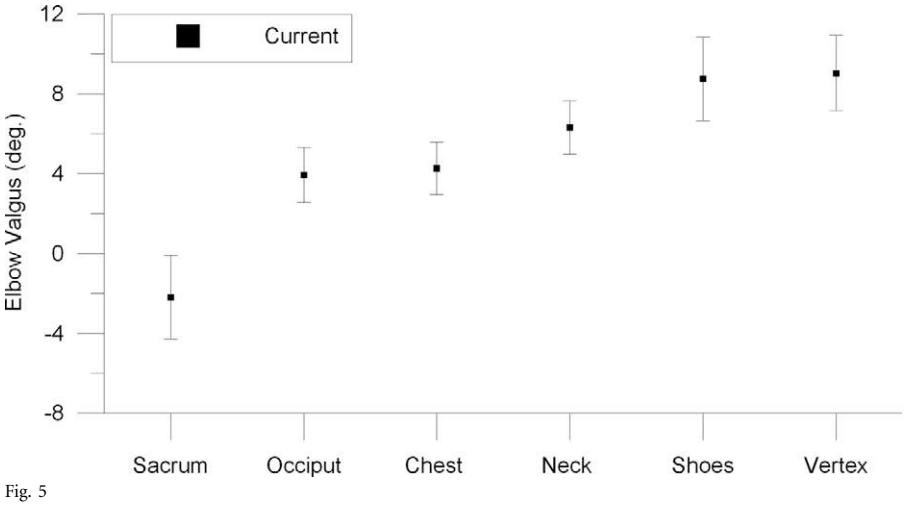


Fig. 5

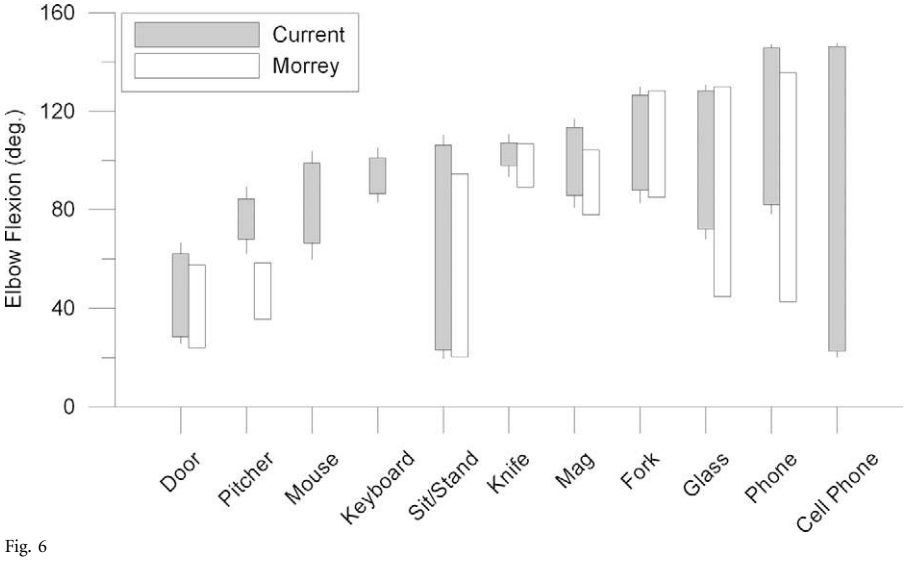


Fig. 6

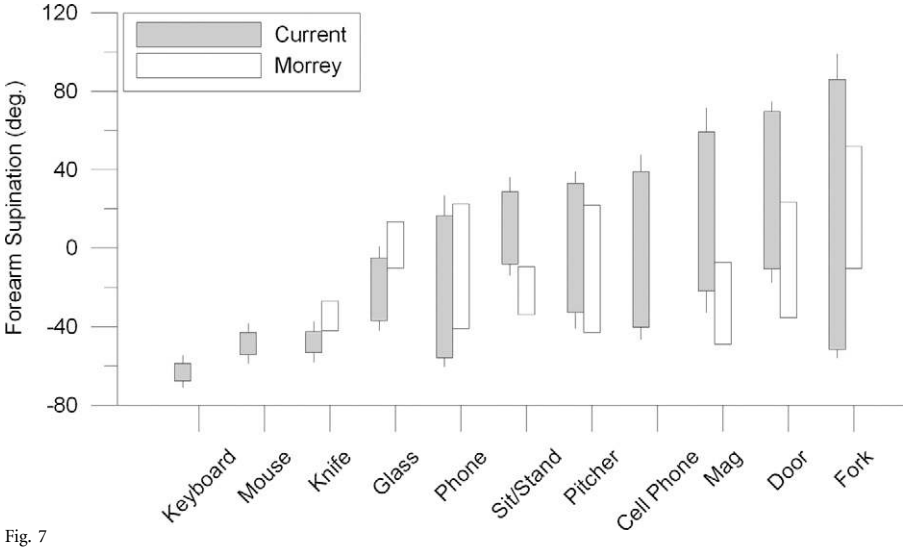


Fig. 7

Fig. 5 Positioning test end points ordered from maximum varus (negative) to maximum valgus (positive). Error bars represent 95% confidence interval. **Fig. 6** Functional task range of motion for elbow flexion ordered by maximum flexion values. Current data are compared with those in the study by Morrey et al.[5]. Error bars represent 95% confidence interval. Mag = reading a magazine. **Fig. 7** Functional task range of motion for forearm rotation ordered from minimum to maximum peak supination. Negative values indicate pronation. Positive values indicate supination. Current data are compared with those in the study by Morrey et al.[5]. Error bars represent 95% confidence interval. Mag = reading a magazine.

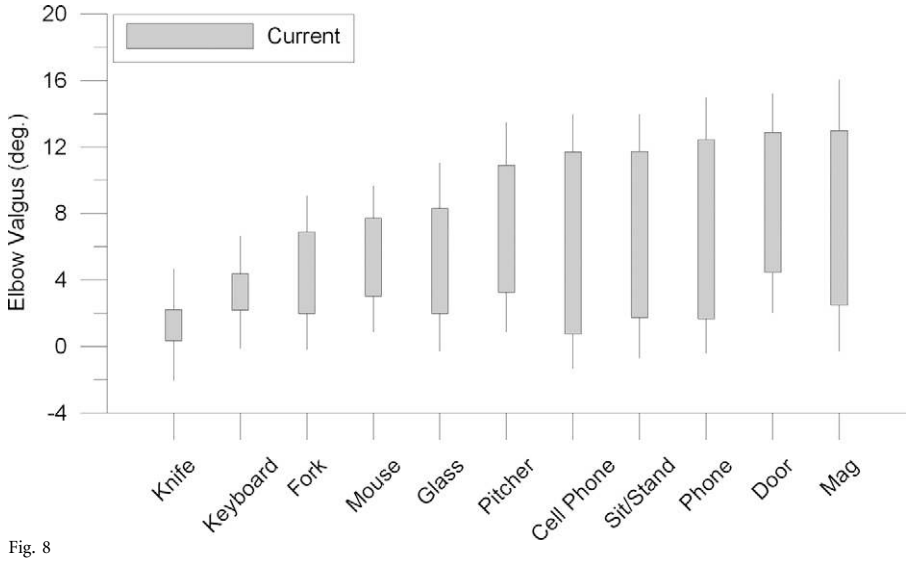


Fig. 8

Functional task range of motion for elbow varus and valgus angulation ordered by peak valgus. Positive values indicate valgus, and negative values indicate varus. Error bars represent 95% confidence interval. Mag = reading a magazine.

and valgus data revealed the largest motion arcs with use of a cellular telephone (11° ± 4°) and rising from a chair (9° ± 3°). Minimum values for valgus were found with using a knife (0° ± 6°), while maximum values for valgus were found with opening a door (13° ± 6°).

Discussion

Functional elbow range of motion on positional and functional tasks has been reported previously by Morrey et al.[5]. With use of a triaxial goniometer, it was determined that 30° to 130° of flexion, 50° of pronation, and 50° of supination were required for personal hygiene and sedentary tasks. These numbers have often been quoted as the standard for functional range of motion about the elbow and have been used to formulate surgical indications regarding elbow stiffness, arthrodesis positioning, and validating outcomes in total elbow arthroplasty.

Despite this report, various other studies that have disputed these findings have emerged over the years[7-10]. With the advent of three-dimensional motion capture technology, which includes electromagnetic and optical tracking methods, several studies have found greater motion arcs for pronation and supination as well as flexion and extension compared with those in the original article by Morrey et al. These models are able to track and measure motion in three different planes simultaneously with a high degree of accuracy, and the measurements are not affected by soft-tissue impingement as potentially seen with the triaxial goniometer.

Safaee-Rad et al.[6], in 1990, were the first, to our knowledge, to report elbow range of motion with use of three-dimensional optical tracking methods. Examining three feeding activities, they found drinking from a cup, eating with a fork, and eating with a spoon all required an arc of flexion from 70° to 130° and 40° of pronation to 60° of supination. These findings were limited by only studying feeding activity, although they were important in that the range of motion confirmed previous data of feeding-related tasks in the study by Morrey et al.

In 2005, Magermans et al.[7] used electromagnetic three-dimensional tracking to evaluate six functional tasks. Electromagnetic tracking and digital optical tracking techniques have been shown to be within 2° of each other in direct comparison studies[15]. Twenty-four female subjects performed simulated tasks of combing their hair, perineal care, eating with a spoon, reaching, washing the axilla, and lifting a 4-kg bag. Flexion values ranged from 61° to 135.7°, and pronation ranged from 71° to 120°. Although flexion values were similar to those found by Morrey et al., there was a marked increase in the amount of pronation required compared with that found by Morrey et al.[5]. In comparison with the results described by Safaee-Rad et al., with use of a three-dimensional optical tracking method, the inclusion of tasks other than feeding markedly increased the required arc of motion[6].

In 2005, standards for defining motion axes were implemented by the International Society for Biomechanics[16]. It is important to understand that three-dimensional motion tracking relies on Euler angles to determine axes of motion and joint centers. The precise definition of joint centers and various axes about them is required if comparisons among studies are to be reliably made. After these guidelines were published, three other studies (one in 2006 and two in 2007) described the range of motion of the upper extremity with use of an optical tracking setup[8-10]. Pieniazek et al.[9] used a system similar to the one used in the present study to evaluate combing one's hair, closing a "zip fastener," and answering a telephone. Eight subjects participated in the study, and maximum flexion values

for picking up a telephone and combing one's hair were reported as 140°. Raiss et al.[10], in a study of eight subjects, used an optical tracking system to evaluate ten activities of daily living. A motion arc for flexion of a mean of 110° (range, 36° to 146°) and a mean pronation-supination arc of 117° (range, 55° of pronation to 72° of supination) were found. Henmi et al.[8] evaluated five subjects for three functional tasks. Eating a meal, shampooing one's hair, and washing one's face were all evaluated with use of an optical tracking system with a similar marker setup. Flexion values ranged from a mean (and standard deviation) of 140° ± 5° for washing one's face to 151° ± 9° for shampooing one's hair. Mean pronation was reported as 75° and mean supination as 100°. These motion arcs and positional values are more consistent with the findings in the present study. Nevertheless, none of those authors replicated all of the tasks evaluated in the study by Morrey et al.[5].

Possible explanations for the differences between the results reported by Morrey et al., with use of the triaxial goniometer, compared with the results with use of optical motion tracking include soft-tissue impingement with the device and/or inadequate alignment of the goniometer along the arm during elbow motion, subsequently altering motion axes. Both of these were identified as potential weaknesses in the original study by Morrey et al.[5]. Three-dimensional motion tracking offers the advantage of not needing to have the subjects to wear a linkage and wires, and potentially allows individuals to perform the tasks more naturally.

Varus and valgus elbow positioning in a normal group of individuals has not been previously reported as far as we know. These data may prove useful, especially when total elbow arthroplasty is considered. The degree to which the varus-valgus motion is coupled with flexion due to axis alignment and Euler angle specifications versus true motion laxity is uncertain. O'Driscoll et al. reported on electromagnetic tracking of eleven cadaveric elbows before and after implantation of a semiconstrained Coonrad-Morrey elbow prosthesis[17]. Varus-valgus laxity was evaluated throughout the flexion arc with the muscles loaded. A mean (and standard deviation) of 6.9° ± 3.7° of varus and valgus laxity was found prior to implantation, while 10.8° ± 1.8° of each was found after implantation. This information suggested that the prosthesis behaved as a "semiconstrained" joint. The authors suggested that because the Coonrad-Morrey prosthesis has more allowable varus-valgus motion than commonly seen in the intact elbow, the implant should never achieve the maximal varus-valgus positions allowed by the design, thereby limiting the development of early wear. That study was the first, as far as we know, to report on the kinematics of the elbow after total elbow arthroplasty, confirming that angular changes are present at the prosthesis with regard to varus and valgus positioning. Our study confirms these in vitro data as we found that, within a normal group of individuals, there is varus and valgus motion of the elbow while performing activities of daily living. Similarly, our in vivo data of 7.0° ± 1.7° of varus and 9.0° ± 2.9° of valgus is close to the 6.9° ± 3.7° determined by O'Driscoll et al. in cadaveric elbows. Further study would include in vivo analysis of varus and valgus motion arcs in patients after undergoing total elbow arthroplasty.

We found supination was much greater than previously reported, particularly for the task of touching the sacrum. While this finding is new and different from the results in the study by Morrey et al., increased supination of greater than 80° or 90° has been previously reported with higher levels of elbow flexion (as in the increased elbow flexion required to touch the sacrum in the present study). Shaaban et al.[18] reported maximum pronation and supination to be greater than previously measured, and related this increase to the flexion and extension position of the elbow. With use of a custom jig and goniometer that fit the arm at the elbow and distal radioulnar joint, fifty volunteers held a grip and moved their forearms from full supination to full pronation with the elbow at full extension, 45° of flexion, 90° of flexion, and full flexion. The forearm supinated fully to 47.4° with the elbow in full extension and to 115.3° with the elbow in full flexion. With greater flexion, there was an increase in supination motion. A possible anatomic explanation for the increased supination with greater amounts of elbow flexion, as reported by Shaaban et al., is secondary to the shape of the proximal radioulnar joint and the known translation of the radial head inside this joint. The proximal aspect of the proximal radioulnar joint is wider than the distal part. The radial head translates volarly with pronation and dorsally with supination as well as distally with extension and proximally with flexion. The amount of forearm rotation depends on the amount of volar-dorsal translation allowed. With flexion, the radial head translates proximally into the wider portion of the joint, allowing greater volar-dorsal translation and, as a consequence, more forearm supination than is typically found at lesser degrees of elbow flexion. This anatomic finding may explain why the maximal supination values of 105° (at flexion angles of >90°) in our study are larger than those typically reported for maximal allowable supination.

The strengths of the present study include the large number of subjects evaluated compared with previous motion analysis studies. With twenty-five subjects, we were able to achieve flexion confidence intervals for each task within 5°. Additionally, a model utilizing a functionally determined elbow axis location and orientation and a glenohumeral joint center may offer improved accuracy over previous methods (as described in our previous study[14]). Another improvement of the current study compared with prior investigations is the fact that we replicated all of the tasks evaluated in Morrey et al., the current standard used to define functional elbow range of motion, with the addition of contemporary tasks not previously evaluated. Finally, the optical tracking system allowed us to determine varus and valgus ranges of motion for subjects with normal elbows, which had not been previously reported.

Limitations of the study can be categorized as those inherent to studying the upper extremity with motion analysis and those specific to our study design and rationale. Unlike gait, upper extremity tasks can be performed in a variety of ways. Repeatability of testing was an issue with this study, as it has been for all previous studies. For example, when a subject is

asked to pick up a pitcher and pour into a glass, there is a great deal of individual variation in the execution of the task, which alters the mean and standard deviation values for the given task. Another disadvantage similar to the variety in execution strategies among individuals is the varying amounts of compensatory motions from the other joints of the spine and upper extremity used to complete a task. Vasen et al. demonstrated this in their study of functional elbow range of motion[19]. They had forty-nine subjects wear a Bledsoe brace, which sequentially limited their range of flexion and extension by 15°, and asked them to perform tasks as outlined by Morrey et al.[5]. They used "completion of the task" as their primary end point and found that subjects could complete most tasks within a range of 75° to 120°. Completion of tasks was facilitated through compensatory motion at other joints. A limitation of any optical motion capture measurement is soft-tissue artifact. Skin motion across the bones, along with movement between markers, is a source of error that cannot be ignored. It is most pronounced in the forearm affecting rotation, but could also affect joint centers and motion axes as markers change position relative to one another through each movement. In this study, a calibration, described in our previous study[14], was applied to adjust for skin motion artifact in supination-pronation. ■

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